

Application Manual

version 2.0

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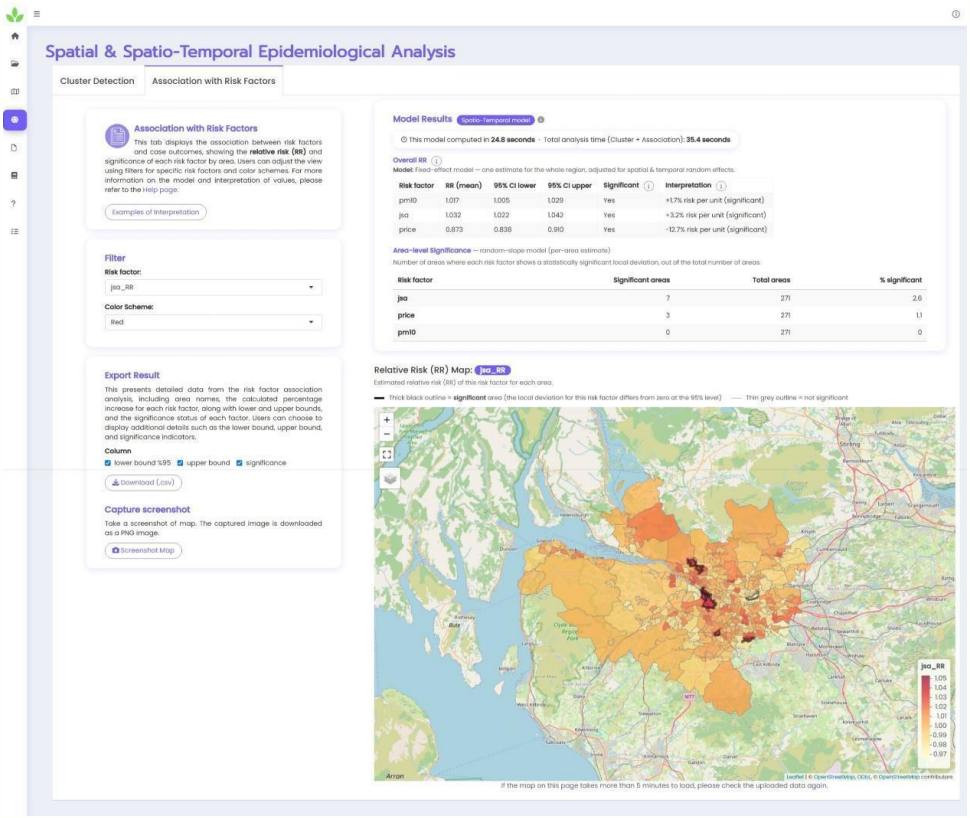
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What is STEHealth?

STEHealth (Spatiotemporal Epidemiological Health) is a browser-based application for the Bayesian analysis of areal health data, supporting disease mapping, hotspot detection, and the estimation of associations between health outcomes and risk factors in both spatial and spatiotemporal settings. Users can analyze their own data or explore the application with built-in sample datasets.



Background

Bayesian spatial and spatiotemporal models are widely used in epidemiology, but fitting them typically requires programming expertise in specialized statistical software, which limits their use by public health practitioners and applied researchers. Existing interactive tools, such as SpatialEpiApp [4], have lowered this barrier, yet options for spatiotemporal risk factor analysis remain limited. STEHealth was developed to bridge this gap by providing an intuitive, browser-based interface built on the RShiny framework, combined with R-INLA for Bayesian hierarchical modeling.

Purpose

To provide an accessible, user-friendly web application that facilitates the visualization and Bayesian analysis of spatial and spatiotemporal epidemiological data.

Features

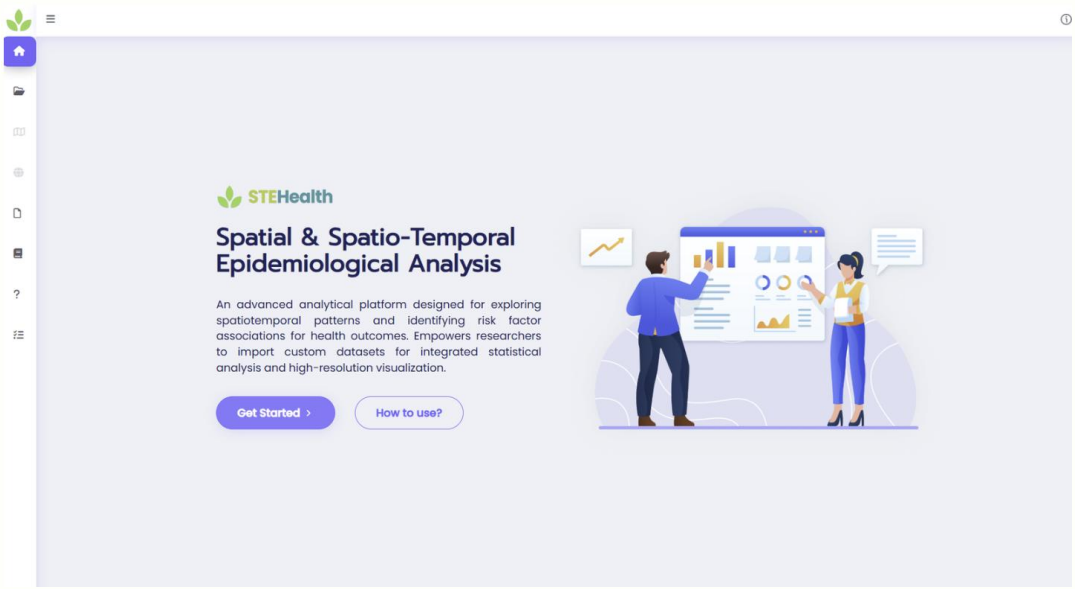
- **Interactive Mapping:** Visualize spatial data using customizable maps, enabling insights into health trends and patterns.
- **Cluster Detection:** Identify hotspots and non-hotspots within your data using Bayesian spatiotemporal models.
- **Risk Factor Analysis:** Explore associations between risk factors and health outcomes, with two complementary views — the area-level relative risk (RR), which lets the effect of a risk factor vary by area, and the Overall RR, a single fixed-effect estimate for the whole study region, directly comparable to region-wide results reported in reference papers.
- **User-Friendly Filters:** Customize visualizations with time points, risk factors, and color schemes.
- **Data Integration:** Upload your own shapefile and case data (CSV or Excel), or use the built-in sample data
- **Ready-to-Use Sample Datasets:** Explore the web application immediately with three built-in sample datasets (spatial and spatial-temporal), complete with a data dictionary for each.

Target Users

- **Primary:** Epidemiologists and public health researchers
- **Secondary:** Public health professionals and policymakers

How to Access

STEHealth is deployed on the cloud via Hugging Face Spaces. **No installations are required.** You can access and use the web application directly through your web browser, from any device (Windows, Mac, Linux, or mobile) with **an internet connection**.



🔗 **Link:** https://mmillorkrn-stehealth-application.hf.space/

Recommendation:

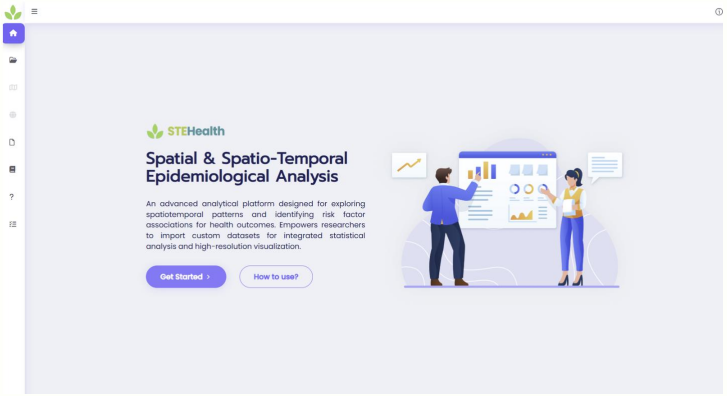
- A modern web browser (Chrome, Edge, Firefox, or Safari, latest version)
- A screen resolution of at least 1440 × 900 pixels on desktop browsers
- A stable internet connection

Note: The web application may enter sleep mode after 48 hours of inactivity to save resources. It will automatically wake up on your next visit, which might take 1-2 minutes to load.

Quick Start: Step-by-Step Demonstration

This section walks through a complete analysis using the built-in sample data. It takes about 10 minutes and requires no data preparation. Each application page is described in more detail in Section 4.

Step 1: Open the web application via the browser link and click the "Get Started" button. No installation is required.

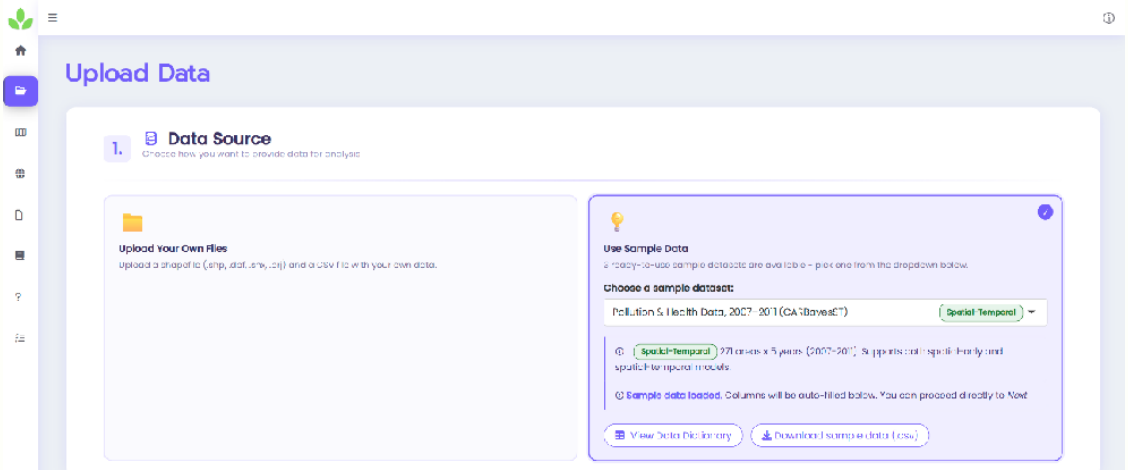


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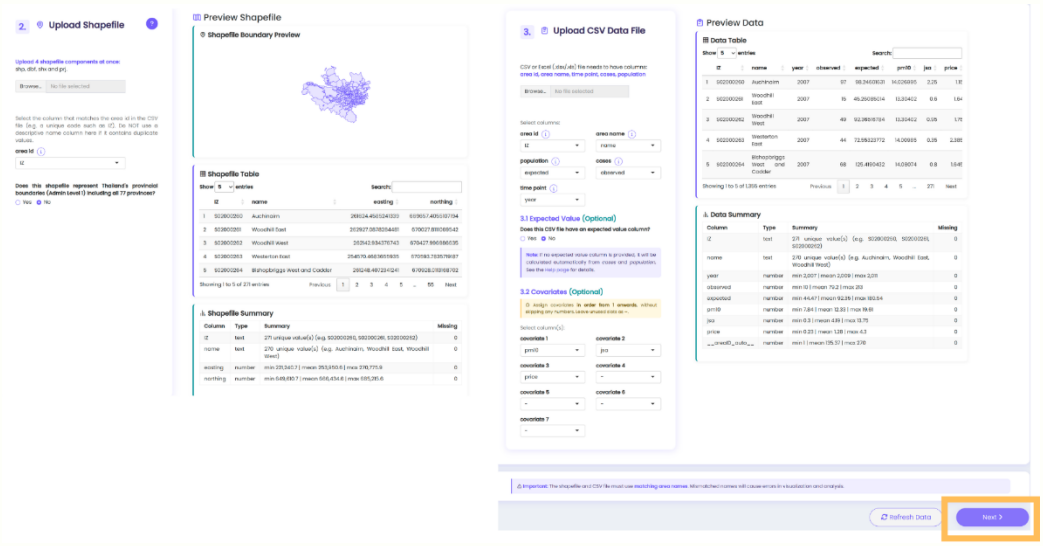
Step 2: Choose one of two paths (for this demonstration, **select path b to use the built-in sample data**):

(a) Use your own data: Upload 4 shapefile components at once (shp, dbf, shx, prj), then select the area id column. Upload a CSV or Excel file and select columns: area id, area name, time point, population, cases.

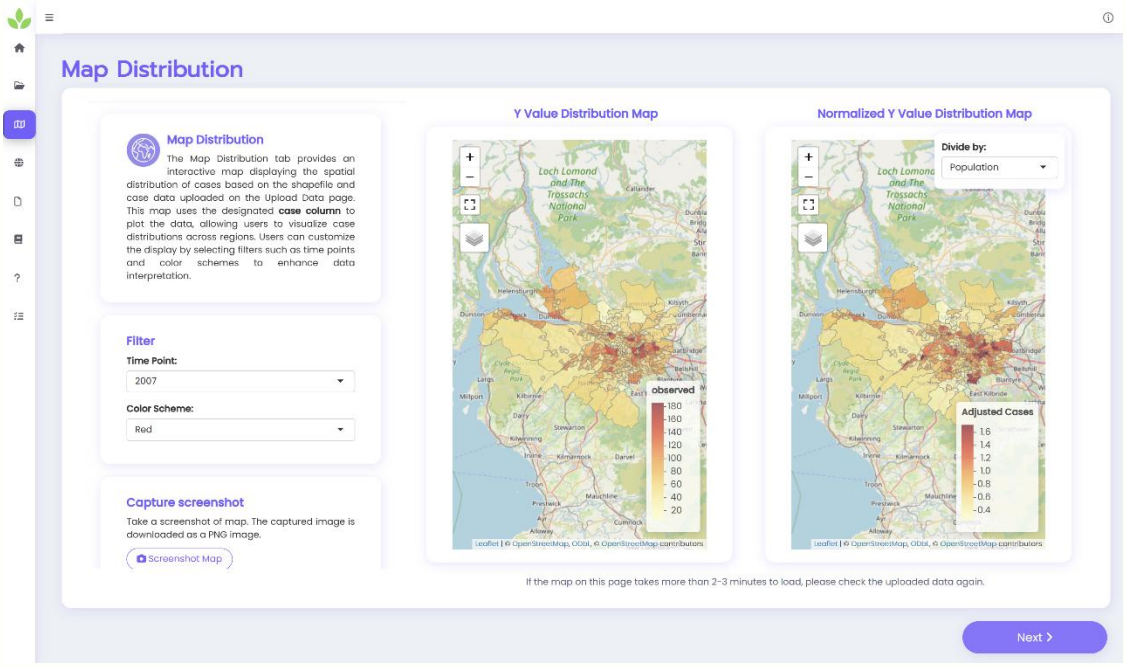
(b) Use Sample Data (recommended for a quick demo): Select Use Sample Data, choose Pollution & Health Data, 2007–2011 (CARBayesST) from the dropdown menu, and optionally check View Data Dictionary first. Once selected, all required shapefile and data file columns will be auto-filled automatically.



Step 3: Review the Data Summary table to verify the dataset, then click **Next** to proceed to the Map Distribution page.

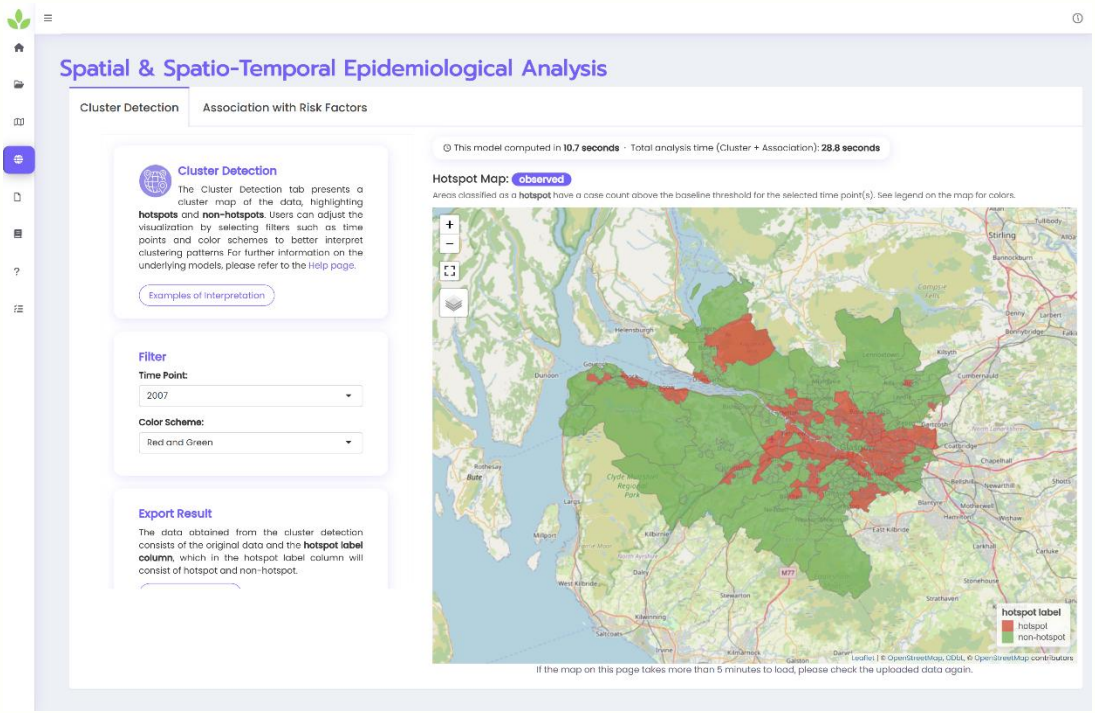


Step 4: On the Map Distribution page, visualize the map using filters (time point, color scheme), optionally take a screenshot, then press **Next** to go to the analysis page.



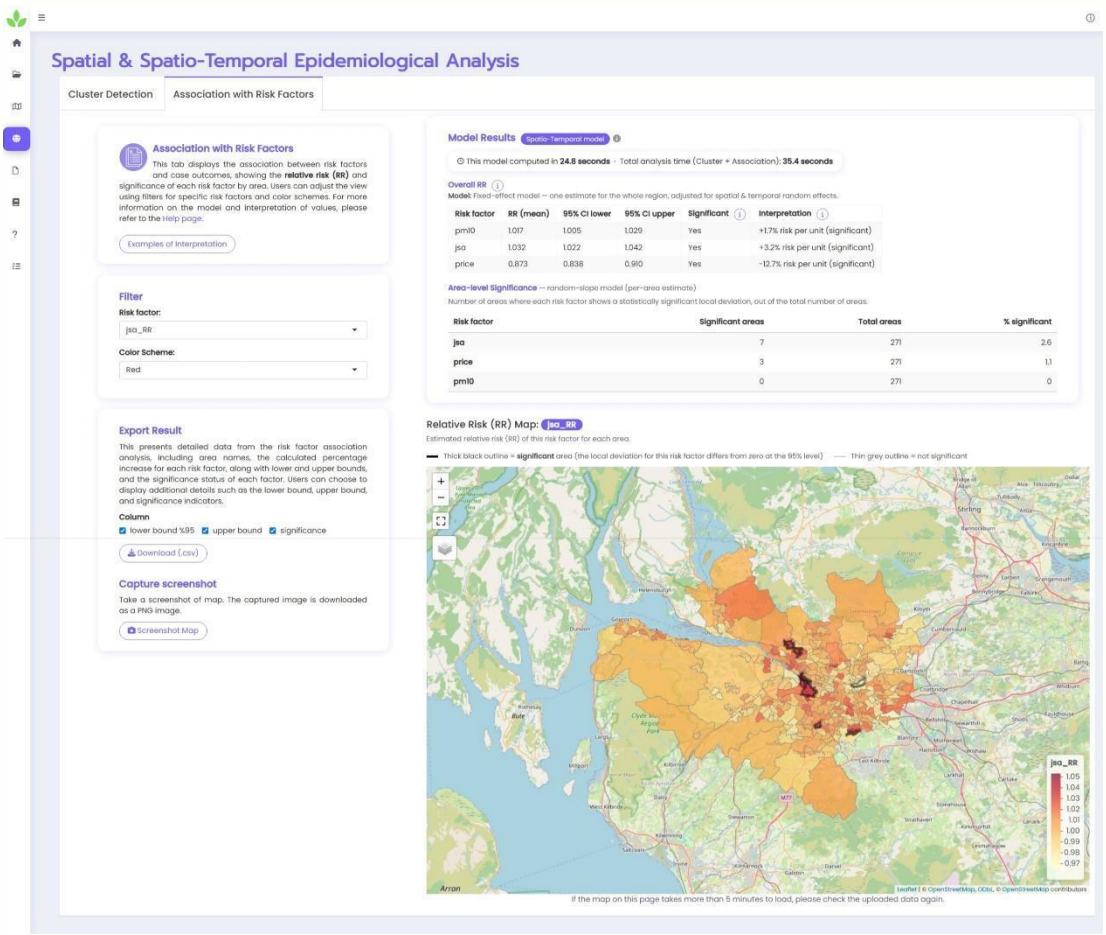
Step 5: **In the Cluster Detection tab:**

- The Cluster Detection tab presents a cluster map of the data, highlighting hotspots and non-hotspots.
- Users can visualize results using filters, download the result data, and take a screenshot.
- For guidance on interpreting the results, users may refer to the Examples of Interpretation button within the web application or Section 5 of this manual.
  Note: On the Spatiotemporal epidemiological analysis page, the analysis may take 1-4 minutes, depending on the data size and current server load.



Step 6: **In the Association with Risk Factors tab:**

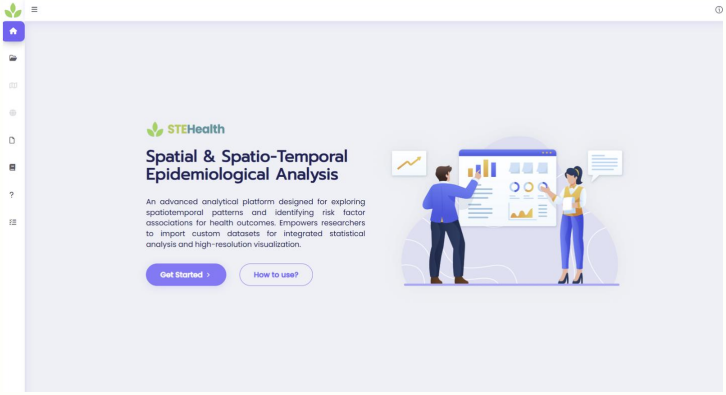
- Scroll to **Overall RR** to examine the region-wide relative risk table for each risk factor, and review the Significant and Interpretation columns.
- Scroll to **Area-level Significance** to observe how many areas show a significant local deviation for each risk factor.
- Use the filters and map to explore area-level relative risk (RR), download results, and take a screenshot.
- For guidance on interpreting the results, users may refer to the Examples of Interpretation button within the web application or Section 5 of this manual.



Web Application Pages

The web application consists of **eight** pages:

1. **Home Page**



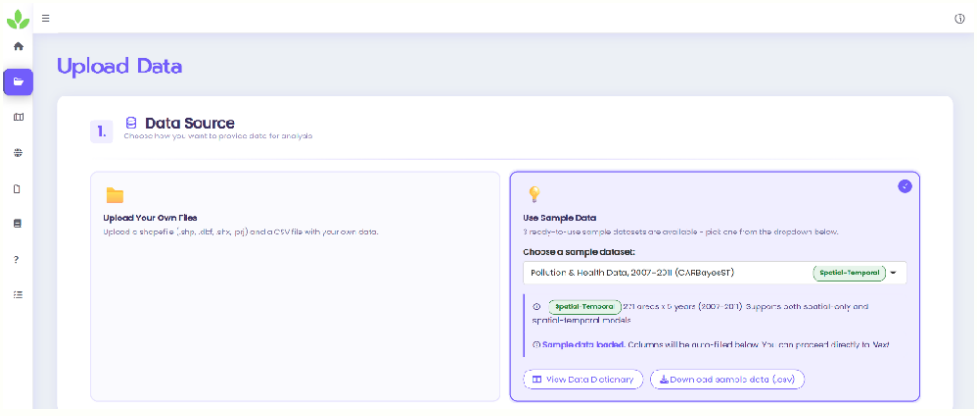
The Home page is the landing page and provides a brief overview of the application. On this page, there are two buttons:

- "Get Started" : Clicking this button takes you to the "Upload Data" page.
- "How to use?": Opens the Manual page, which provides step-by-step instructions and a worked demonstration of the application.

2. **Upload Data Page**

The Upload Data page allows users to upload data to be analyzed, or get started instantly with sample data. The data importation process on this page is divided into three main sections:

2.1. Data Source



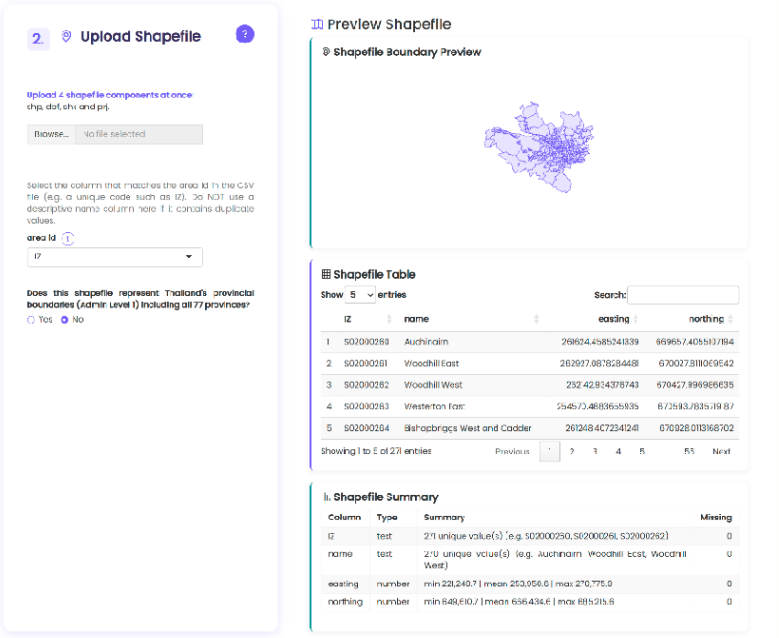
In this section, users must select the method for providing data for the analysis. There are two primary options:

- **Upload Your Own Files:** Select this option to import your own spatial data (Shapefile) and tabular data (CSV or Excel) for analysis.

- **Use Sample Data:** Instead of uploading their own files, users can select this option and pick one of three ready-to-use sample datasets from a dropdown. The available sample datasets include:
    - Pollution & Health Data 2007-2011
    - Pollution & Health Data, 2007 Only

2.2. Upload Shapefile

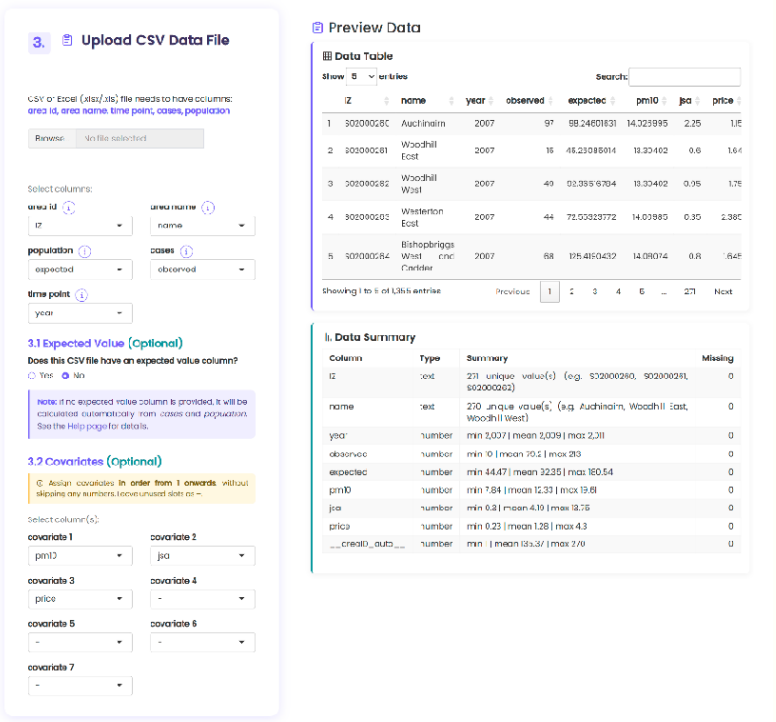
If users choose to upload their own files, they must upload the spatial data (Shapefile) in this section. Users can upload their own shapefile for any region



- **File Upload:** Click the "Browse..." button to upload the spatial data. Shapefile upload allows users to **upload all 4 files at once**: shp, dbf, shx, and prj.
- **Column Selection:** Select the area id column from the dropdown menu. The selected column must match the area id column in the CSV/Excel file.
- **Boundary Confirmation:** Select "Yes" or "No" to specify whether the uploaded shapefile represents Thailand's provincial boundaries (Admin Level 1) including all 77 provinces.
- **Preview Shapefile:** Upon successful upload, a visual preview of the spatial boundaries will be displayed in the panel on the right.

2.3. Upload Data File (CSV or Excel)

In this final section, users upload the dataset containing epidemiological outcomes and demographic information.



- **File Upload:** Click the "Browse..." button to select the data file. The user can upload one data file in CSV or Excel (.xlsx/.xls) format.
- **Variable Mapping:** Users are required to map the columns in their uploaded dataset to the system's required variables using the dropdown menus. The file must contain the following variables: area id, area name (used as the map label), cases, time point, and population.

2.3.1. **Expected Value** (Optional)

- Users must select "Yes" or "No" to indicate if the uploaded file contains an expected value column.
- If the uploaded file lacks an expected value column, the system will automatically calculate it from the cases and population data using the following formula:

$$Expected\ Value = \frac{sum(cases)}{sum(population)} * population$$

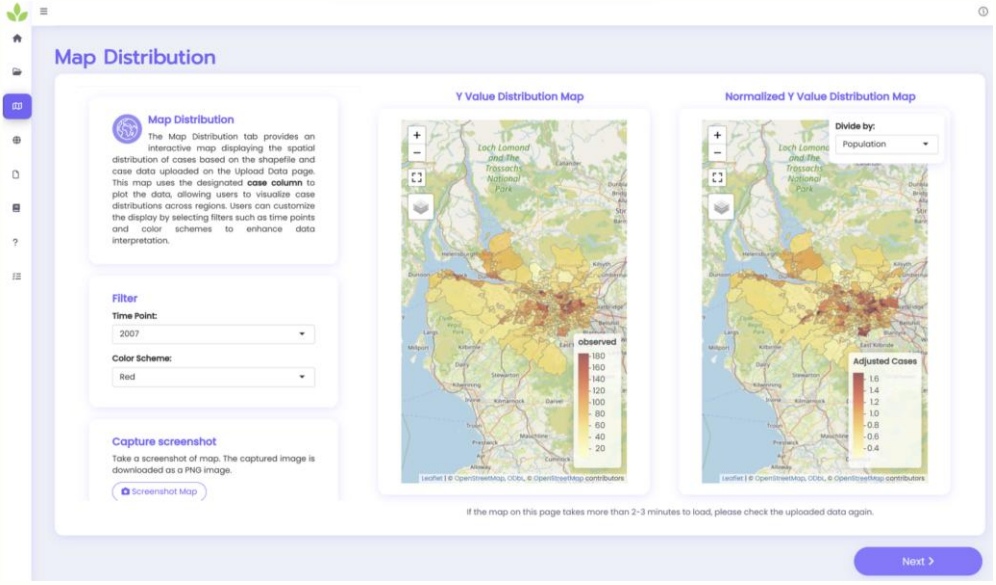
- If it has one, the user selects it from a dropdown.

2.3.2. **Covariates** (Optional)

- If performing a risk factor analysis, users can assign covariates in this subsection.
- Arrange covariates in sequential order from 1 to 7 with no gaps.
- Selecting 1 covariate performs a univariate analysis; selecting more than 1 performs a multivariable analysis.
- Note: The RR is interpreted per one-unit increase in the risk factor, based on the scale and unit of the uploaded data.
- Preview Data: Once the data file is successfully loaded, a data summary table will be displayed in the preview panel on the right.

Important: Please ensure the uploaded shapefile matches the data file, with consistent area names/codes. Mismatched files may lead to errors in visualization and analysis.

3. Map Distribution Page



The Map Distribution page provides an interactive map displaying the spatial distribution of cases based on the shapefile and case data uploaded on the Upload Data page. This map uses the designated case column to plot the data, allowing users to visualize case distributions across regions. Users can customize the display by selecting filters such as time points and color schemes to enhance data interpretation.

Control Panel

- **Filter:** Users can select a specific Time Point and Color Scheme from the dropdown menus to update the map visualizations.
- **Capture Screenshot:** Users can take a screenshot of the map by pressing the Screenshot Map button. The captured image is downloaded to your device as a PNG image**.**

Map Visualizations

The main interface displays two side-by-side maps:

- **Y Value Distribution Map:** Displays the spatial distribution of the raw, observed cases.
- **Normalized Y Value Distribution Map:** Displays adjusted cases and includes a "Divide by:" dropdown menu (e.g., dividing by Population) to normalize the data for comparative analysis.

Interactive Features

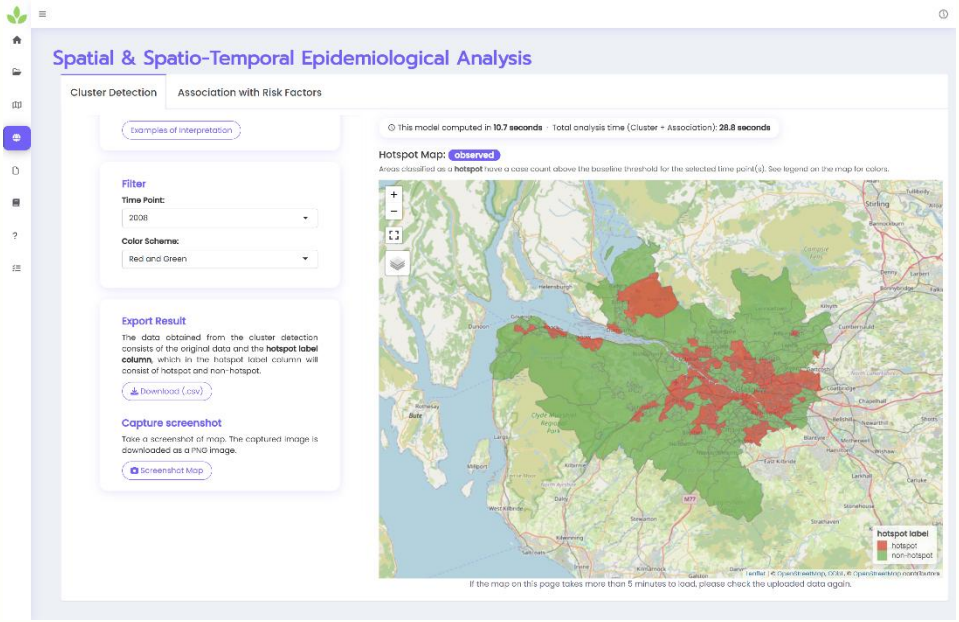
For both map displays, users can interact with the spatial data using the following controls:

- Hover over an area to display the specific value of that area.
- Zoom in or zoom out by pressing the + or - buttons.
- View the map in fullscreen mode by pressing the fullscreen icon.
- Change the map theme by pressing the layers icon.

4. Spatial & Spatio-Temporal Epidemiological Analysis

This page displays the model results once data have been uploaded. The application automatically detects whether the data contain more than one time point, fits a spatiotemporal or a spatial model accordingly, and indicates the model type with a badge above the Model Results. This page has two tabs:

4.1. Cluster Detection tab



This tab is divided into two primary sections:

1. Control Panel (Left Sidebar)

- Description: Provides a brief overview and an interpretation guide.
- Filter: Allows users to adjust the map's Time Point and Color Scheme.

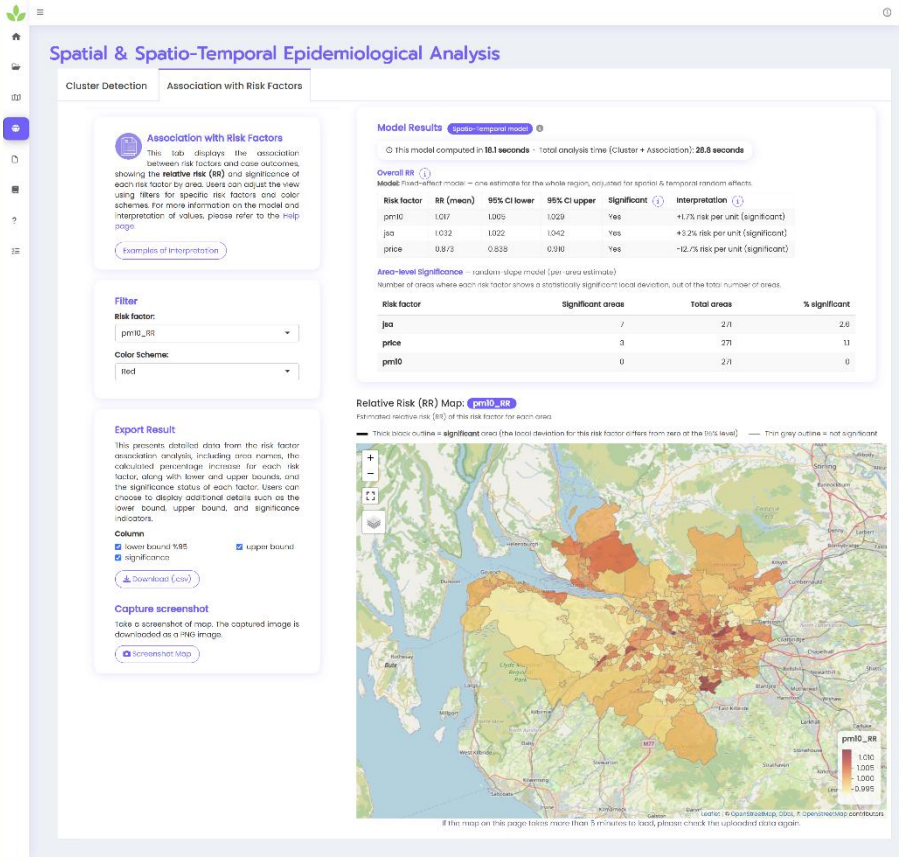
- Export Result: Provides options to download the cluster detection data.

2. Analysis Results Panel (Right Workspace)

- Computation Time: Displays the total model processing time.
- Hotspot Map: An interactive map categorizing and highlighting areas as either hotspots or non-hotspots.

Cluster detection is an important tool for identifying areas of high risk and developing hypotheses about health outcomes [1]. In a Bayesian framework, the posterior distribution of the relative risk can be used to compute the probability that the risk in each area exceeds a given threshold [2]. This exceedance probability is then used to classify each area as a hotspot or non-hotspot [3]. The Cluster Detection tab displays the resulting **hotspot map**. A worked example of how to read this map is given in Section 5, and the underlying model is described in Section 6.

4.2. Association with Risk Factors tab



This tab is divided into two primary sections:

1. Control Panel (Left Sidebar)

- **Description:** Provides a brief overview and an interpretation guide.
- **Filter:** Allows users to select a specific Risk factor and map Color Scheme.
- **Export Result**: Provides options to download analysis data as a .csv file and capture map screenshots.

2. Analysis Results Panel (Right Workspace)

- **Computation Time:** Displays the total model processing time.

- **Model Results:** Displays computation time alongside two statistical tables (Overall RR and Area-level Significance).
- **Relative Risk (RR) Map:** An interactive map visualizing the estimated relative risk for the selected risk factor.

Model Results presents two complementary sets of estimates, each derived from a different model:

1. **Overall RR**

Overall RR

Model: Fixed-effect model — one estimate for the whole region, adjusted for spatial & temporal random effects.

Risk factor	RR (mean)	95% CI lower	95% CI upper	Significant	Interpretation
pm10	1.017	1.005	1.029	Yes	+1.7% risk per unit (significant)
jsa	1.032	1.022	1.042	Yes	+3.2% risk per unit (significant)
price	0.873	0.838	0.910	Yes	-12.7% risk per unit (significant)

- **Overview:** Represents the overall (region-wide) relative risk for each risk factor.
- **Model Type:** Estimated using a fixed-effect model, which provides a single Relative Risk (RR) shared across the whole study region.
- **Statistical Values:** The results table displays the posterior mean RR alongside the lower and upper bounds of its 95% credible interval (CrI).
- **Significance Column:** Indicates "Yes" or "No" based on whether the 95% CrI excludes RR = 1.
- **Interpretation Column:** Provides a plain-language explanation of the data (e.g., "+3.2% risk per unit (significant)").
- **Tooltips:** Users can hover over the (i) icons to view specific details on how each column is calculated.

2. **Area-level Significance**

Risk factor	Significant areas	Total areas	% significant
jsa	7	271	2.6
price	3	271	1.1
pm10	0	271	0

- **Overview:** Represents the area-specific relative risk for each risk factor.
- **Model Type:** Utilizes a random-slope model that allows the effect of the risk factor to vary by area.
- **Results Display:** Shows the number of areas where each risk factor is statistically significant, out of the total number of areas.
- **Significance Criterion:** An area is considered statistically significant when the 95% credible interval (CrI) of its area-specific RR does not include RR = 1. The same significance criterion is applied to both Overall RR and Area-level RR.

Overall RR describes the association between a risk factor and the outcome across the whole study region, whereas Area-level RR shows how this association varies across individual areas.

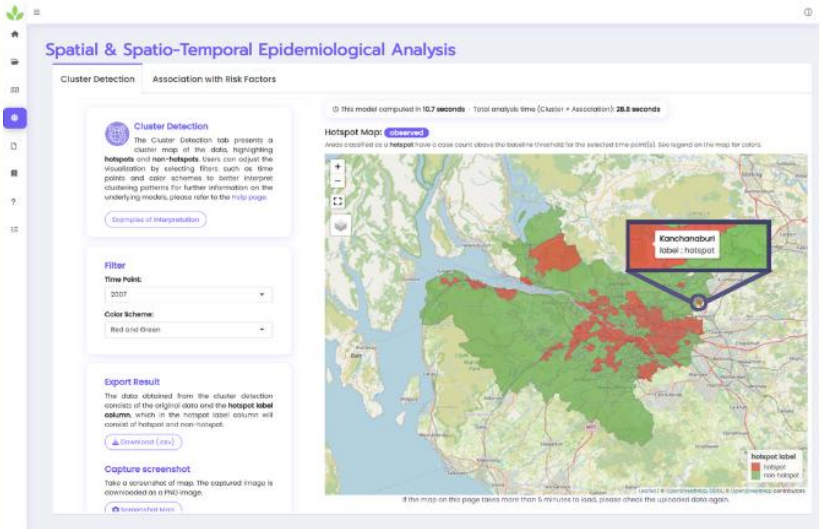
These are two separate models answering two different questions — a risk factor can be significant overall while showing few or no significant areas on the map, or vice versa; this is expected, not an error.

The relative risk (RR) map supports the same filters, hover tooltips, and zoom, fullscreen, and theme controls described for the Map Distribution page. Results can be exported as a .csv file in long (tidy) format, with one row per area per risk factor. Worked examples of how to interpret these results are given in Section 5, and the model specifications are given in Section 6.

Interpreting the Results

This section explains how to read the output of STEHealth using the built-in Pollution & Health sample data. It is written for users without a statistical background; the models behind these results are documented in Section 6. Key terms. Relative risk (RR): the risk in an area, or per one-unit increase in a risk factor, compared with the study-region average. RR = 1 means no difference, RR > 1 means higher risk, and RR < 1 means lower risk. 95% credible interval (CrI): the range that contains the true RR with 95% probability, given the data and the model. Hotspot: an area with a high probability that its RR is greater than 1.

Interpreting the Hotspot Map (Cluster Detection)



In 2007, Moodiesburn West was classified as a hotspot, meaning that the posterior probability that its relative risk of respiratory hospital admissions exceeded the baseline was above the specified threshold (the baseline, RR = 1, corresponds to the average risk across the study region).

Interpreting Risk Factor Associations (Multivariable Example)

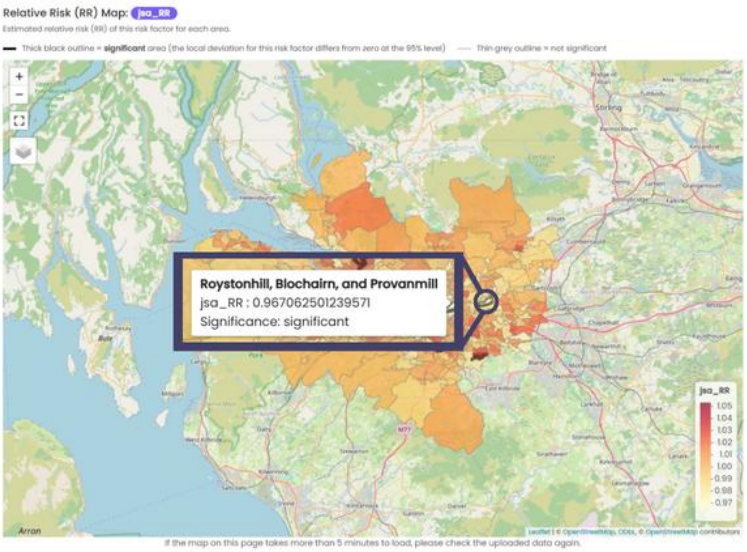
This example interprets a multivariable analysis of respiratory hospital admissions in Greater Glasgow, Scotland, using the built-in Pollution & Health sample data (2007–2011). Three risk factors were entered into the model simultaneously; results are displayed for one risk factor at a time, and each RR is adjusted for the other covariates in the model.

- If the area is **significant**
    - If the **RR value is > 1**



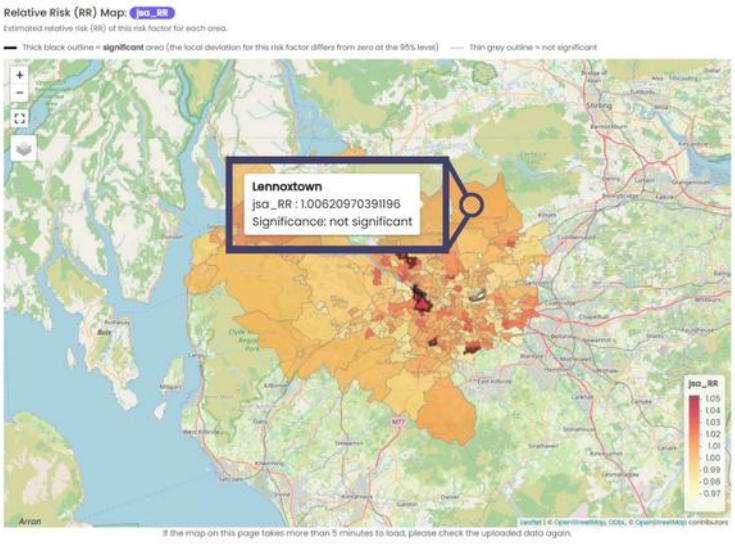
In Carnwadric West, the relative risk (RR) of respiratory hospital admissions associated with the proportion of working-age residents claiming Job Seekers Allowance (JSA) is 1.0391. This suggests that for every 1-percentage-point increase in the JSA proportion, the risk of respiratory hospital admissions increases by 3.91%. Because the RR is multiplicative, larger differences compound; for example, a 10-percentage-point higher JSA proportion corresponds to an RR of approximately 1.47 (1.0391^10).

- If the **RR value is < 1**



In Roystonhill, Blochairn, and Provanmill, the relative risk (RR) of respiratory hospital admissions associated with the JSA proportion is 0.9671. This indicates that for every 1-percentage-point increase in the JSA proportion, the risk of respiratory hospital admissions decreases by 3.29%. For a 10-percentage-point higher JSA proportion, the corresponding RR is approximately 0.72 (0.9671^10). Such locally inverse associations should be interpreted cautiously, as they may reflect residual confounding or small-area instability.

- If the area is **not significant**



In Lennoxtown, the 95% CrI of the area-specific RR for JSA includes 1; therefore, there is no evidence of an association between the JSA proportion and respiratory hospital admissions in this area.

Note: The observed association between a risk factor and health outcome does not necessarily imply a causal relationship.

Statistical Methods (Technical Details)

This section documents the statistical models implemented in STEHealth for transparency and reproducibility. Users who only wish to run analyses and interpret the output may skip it. All models are fitted in a Bayesian framework using integrated nested Laplace approximation (INLA) via the R-INLA package [8].

Cluster Detection Model

STEHealth performs cluster detection using a Bayesian spatiotemporal model. Let y(it) denote the observed count of the health outcome (e.g., deaths or hospital admissions) in area i at time t. Following standard Bayesian disease mapping practice, y(it) is modeled as a function of an intercept and spatial, temporal, and space–time random effects:

$$y_{it} \sim Poisson(E_{it} p_{it}),$$
$$\log(p_{it}) = \alpha + \eta_{it}$$

The random effects capture variation in risk across areas and over time. The spatial effect follows the Besag–York–Mollié (BYM) specification [6], which combines a spatially structured component with an intrinsic conditional autoregressive (ICAR) prior and an unstructured, independent and identically distributed (i.i.d.) component:

$$u_i \mid u_{-i} \sim Normal\left(\mu_i + \frac{1}{N_i}\sum_{j=1}^{n} a_{ij}(u_j - \mu_j), s_i^2\right) + u_i \sim iid\ ormal0, \sigma^2$$

The structured temporal effect is modeled with a first-order random walk (RW1) prior:

$$\gamma_t \mid \gamma_{t-1} \sim Normal(\gamma_{t-1}, \sigma^2)$$

The space–time interaction follows the Type I specification of Knorr-Held [7]. In this specification, the unstructured spatial and unstructured temporal effects interact, so the interaction terms are assigned independent and identically distributed priors across all area–time combinations.

Risk Factor Models

- **Model notation**

i = area/location
t = time point
y_{it} = observed number of cases
E_{it} = expected number of cases
ρ_{it} = relative risk
α = intercept
x_{mit} = risk factor m
β_m = coefficient of risk factor m

- **Fixed-Effect Model (Overall RR)**

  In addition to the area-level analysis, overall relative risk (Overall RR) is estimated using a fixed-effect model to evaluate region-wide associations. Let y_{it} denote the observed counts of the health outcome for location i and time t. The model is structured as a combination of intercept and fixed effects (risk factors) across the whole study region:

$$y_{it} \sim Poisson(E_{it}\rho_{it})$$

$$\log(\rho_{it}) = \alpha + \sum_{m}^{M} \beta_m x_{mit}$$

  The risk factor association is modeled on the multiplicative scale, providing a single fixed-effect estimate for the entire study region, which is directly comparable to region-wide results reported in reference papers.

- **Area-level Model (Random-Slope Model)**

  The per-area relative risk analysis utilizes a Bayesian spatiotemporal model to estimate the effect of risk factors for each specific area. Let y_{it} be the observed counts of the health outcome for location i and time t. The model is formulated as a combination of an intercept, fixed effects (risk factors), and random effects:

$$y_{it} \sim Poisson(E_{it}\rho_{it})$$

$$\log(\rho_{it}) = \alpha + \sum_{m}^{M} \beta_m x_{mit} + \eta_{it}$$

  The risk factor association is modeled on the multiplicative scale, allowing the effect of a risk factor to vary by area. For a comprehensive explanation of the spatial effect, temporal effect, and space-time interaction components in this model, please refer to the Cluster Detection Model section above.

Source code, Dependencies, References, Credits, and Contact us

Source code

The source code for STEHealth is available on GitHub.

Dependencies

- **Deployment & Infrastructure**

Infrastructure	Description
Hugging Face Spaces	Cloud platform for hosting the machine learning and Shiny web application.
Docker	Containerization technology used to create a reproducible environment and manage complex spatial/system dependencies.

- **Software & R Packages**

R packages	Description
R	Language and environment for statistical computing and graphics (version 4.5 or later).
shiny	Makes it easy to build interactive web apps straight from R.
shinydashboard	Use with Shiny to create dashboards.
ShinydashboardPlus	Extend shinydashboard with AdminLTE2 components.
shinyjs	Perform common JavaScript operations in Shiny apps.
shinyBS	Adds additional Twitter Bootstrap components to Shiny.
shinyWidgets	Collection of custom input controls and UI components for Shiny applications.
bsplus	Adds Bootstrap-powered JavaScript functionality (e.g. tooltips, popovers) to Shiny apps.
dplyr	A fast, consistent tool for working with data frame-like objects.
ggplot2	Creates elegant data visualisations using the grammar of graphics.
leaflet	Create Interactive Web Maps with the JavaScript 'Leaflet' Library.
leaflet.extras	Adds extra functionality to leaflet via additional Leaflet plugins.
RColorBrewer	Provides color schemes for maps and other graphics.
sf	Simple Features: modern support for spatial vector data, used for shapefile handling (replaces the legacy rgdal).
spdep	Spatial Dependence: Weighting Schemes, Statistics.
R-INLA	Performs full Bayesian analysis on generalised additive mixed models using Integrated Nested Laplace Approximations. The application uses the INLA testing build [8].
readxl	Reads Excel (.xls/.xlsx) files, used for the Excel upload option on the Upload Data page.
DT	Renders interactive data tables (e.g. the area-level significance table on the Association with Risk Factors tab).
capture	Add a button in Shiny application to take a screenshot of a specified element.

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Credits

- logo of STEHealth web application was modified from logo by Ally Hamid on Canva
- Image in Home page created by pikisuperstar on Freepik
- Images in Upload Data page, Spatial & Spatio-Temporal Epidemiological Analysis page, and About Web Application created by Katerina Limpitsouni on undraw
- World icons in Spatiotemporal Epidemiological Analysis Result page created by Freepik on Flaticon
- World map icons in Spatiotemporal Epidemiological Analysis Result page created by Freepik on Flaticon
- Document icons in Spatiotemporal Epidemiological Analysis Result page created by smalllikeart on Flaticon
- Manual cover in STEHealth web application was modified from report cover by Temptackle on Canva

Contact us

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